

weights. Thus the average investor collects no dynamic factor risk premiums because Mr. Market holds all securities and does not continuously trade.

If you are average, then you're done! And if you use cheap index funds, you'll outperform two-thirds of active managers (see chapter 16). What the mean-variance framework and the equilibrium Capital Asset Pricing Model (CAPM) get spot-on right is that the market portfolio is well diversified; in fact, the market is the most diversified portfolio you can passively construct.

If you are different from average, then you will optimally *NOT* hold market weights. How you tilt away from the market will depend on your investor-specific characteristics. The important question is: "How are you different from average?" Do you have a longer horizon than the typical investor, as Norway does? Are you terrified of highly volatile periods because you need to meet a fixed liability stream? Do you tremble when the market crashes because your income is directly linked to market performance, so you get double-whacked as your financial wealth comes crashing down just as you lose your job?

Each factor defines a different set of bad times. The investor should ask, for example, whether the bad times defined by the value-growth factor are bad times for me? And if they are bad for me, do they hurt less than for a typical market participant?

There are three steps, then, in the factor allocation process:

1. How am I different from average?

Identify your comparative advantages. The Professors' Report enumerated these for the Norwegian fund: (1) transparency and a long-term mandate, (2) large size, (3) long-term horizon and the lack of immediate cash liabilities, and (4) the fund manager NBIM is capable, has a strong public service ethos, and mitigates the problem of agency which is the central challenge of delegated investment management.

Identify your comparative disadvantages. Size is a two-edged sword for Norway, and being large means small-scale alpha opportunities do almost nothing for the overall portfolio. If you're an endowment, perhaps you can't afford big losses because your university will have trouble meeting payroll (this happened to Harvard, see Chapter 13).²⁴ Or do you tend to go-with-the-flow and lavish attention on the latest hot stock?

2. What losses during bad times can I bear?

In each factor's definition of bad times, assess your risk-bearing capacity. This is essentially your risk aversion with respect to each factor. If you are so risk-averse you can't afford any losses during a factor's set of bad times, then you should take the opposite positions in those factors. That is, perhaps you should be doing growth instead of value, or buying volatility protection instead of selling it.

²⁴ See "Liquidating Harvard," Columbia CaseWorks ID #100312.